

Device-Guided Stanford RRAM Model Extraction for Sputter-Deposited TaO_x Devices

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Summary and Motivation

We calibrated the Stanford–PKU RRAM compact model to measured DC switching data from sputter-deposited TaO_x devices fabricated under twelve oxygen flow and RF sputter-power conditions. The goal was not only to obtain a good simulated I – V overlay. For a semiconductor modeling and fabrication group, the more important question is: *which fitted parameters can be trusted as deposition-dependent information, and which are only numerical knobs that help the curve fit?*

This distinction matters because the Stanford model has many adjustable parameters, while a DC butterfly sweep contains only a limited number of visible features: the low- and high-resistance branches, SET and RESET transition voltages, branch curvature, and current compliance region. Many parameter combinations can reproduce these features. Therefore, a good overlay is necessary for circuit use, but it does not automatically prove that each extracted parameter has physical meaning.

Our extraction flow adds two simple checks after fitting. First, we test whether a model fitted to one representative cycle predicts all other measured cycles of the same device. This tells us whether the fit represents the device population rather than only one selected trace. Second, we repeatedly resample the measured cycles and refit the model. This is called a bootstrap; here it simply means “refit many slightly different cycle populations and see which parameters stay stable.” A stable parameter is useful for interpretation. A parameter that moves widely is treated as a fitting degree of freedom, not a process metric.

Device Set and Methodology

The devices are sputter-deposited TaO_x cross-point RRAM cells with 4 μm feature size. Twelve sample conditions span oxygen percentage in the sputter ambient and RF sputter power, including four replicated center-point conditions. This design allows deposition trends to be checked, but only for parameters that are repeatably extracted.

The extraction flow has four main steps. First, measured cycles are screened to reject shorted, open, incomplete, or noise-floor sweeps. Second, one representative measured cycle is selected automatically. We use the *medoid*, which is the real measured cycle closest to the median cycle behavior. This avoids hand-picking a visually attractive curve and also avoids fitting an artificial averaged loop.

Third, before compact-model fitting, each resistance-state segment is classified as ohmic, Poole–Frenkel, Schottky, or Fowler–Nordheim. This is a device-physics front end. Ohmic and Poole–Frenkel-like regions are closest to what the released Stanford current law can represent. Schottky and Fowler–Nordheim regions are still reported and validated, but they are treated as possible model-mismatch regions during fitting. This prevents interface-limited HRS behavior from forcing unrelated bulk-model parameters to compensate.

Fourth, the Stanford Verilog-A model is fitted using HSPICE simulations. The objective compares measured and simulated currents on a log-current scale, because the current spans many decades. The search is guided by physically reasonable parameter ranges and by the conduction classification. In plain language, the algorithm searches broadly enough to fit the data, but it is discouraged from using unsupported parameters to hide missing physics.

Results and Interpretation

The first result is a device-physics result. The post-SET low-resistance state is consistently ohmic, as expected for a conductive filament branch. The post-RESET high-resistance state (HRS) is Schottky-dominant in 92.5–100% of cycles for all twelve conditions. The extracted dynamic permittivity values are physically admissible, supporting the conclusion that the HRS is interface-limited rather than simply noisy or poorly fitted. This also explains why RESET remains harder to fit: the released Stanford model does not explicitly include a Schottky barrier current.

The fitted models reproduce the measured DC switching loops with mean log-current RMSE of 0.530 decades for SET and 0.761 decades for RESET. The dominant residuals occur near the switching transition and along the Schottky-limited HRS approach. Thus, the remaining error is mainly a model-adequacy issue, not a failure of the optimizer.

The out-of-sample cycle test shows that the model generalizes. When the parameter set fitted to one medoid cycle is scored against every other measured cycle of the same device, the held-out median errors are 0.542 decades for SET and 0.800 decades for RESET. These are very close to the fitted-cycle errors. Therefore, the extracted model is a behavioral description of the device cycle population, not just a memorized overlay of one trace.

The bootstrap analysis shows which parameters can be interpreted. The branch voltage scales are the most repeatable quantities and carry meaningful deposition trends. This is reasonable because these voltage scales directly control branch curvature in the measured I – V curve. In contrast, series resistance and velocity prefactors vary strongly when the measured cycles are resampled. They may be needed to produce a good fit, but they should not be reported as standalone deposition-dependent physical parameters from these DC sweeps.

Novel Contributions

The work makes four contributions. First, it reframes Stanford-model calibration as a device-facing extraction problem: the deliverable is not only a curve overlay, but also evidence that the fit generalizes and that selected parameters are repeatable. Second, it places conduction-regime classification before fitting, so the method knows where the TaO_x data follow the compact model and where interface-limited HRS transport lies outside it. Third, it shows that a single medoid-cycle calibration predicts the full measured cycle population of the same device. Fourth, it separates deposition-useful parameters from fit-only parameters: branch voltage scales are usable for process comparison, while series resistance, velocity prefactors, activation energies, and several gap-related quantities are not repeatable enough for physical interpretation from DC data alone.

Conclusion

For these sputter-deposited TaO_x RRAM devices, a good compact-model overlay is only the first checkpoint. The fitted Stanford model is useful as a behavioral model because it predicts held-out measured cycles with nearly the same error as the fitted cycle. However, the extracted parameter table must be interpreted selectively. Only repeatable quantities, especially the branch voltage scales, should be used to discuss deposition dependence. The post-RESET HRS is consistently Schottky-limited, which sets the RESET error floor and points to a clear model-extension direction: adding an interface-limited HRS current branch. The central message is that RRAM model extraction for fabrication studies should report fit quality, missing device physics, cycle-level generalization, and parameter repeatability together.

Figures

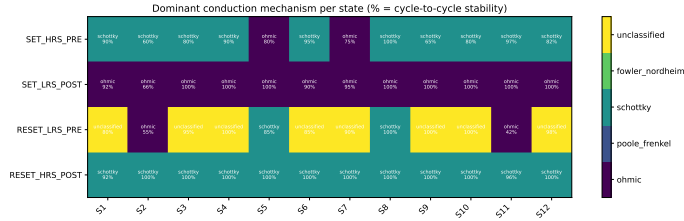


Figure 1: Dominant conduction mechanism for each state segment and deposition condition. The main pattern is the device story: post-SET LRS is mostly ohmic, while post-RESET HRS is Schottky-like for essentially all conditions. This identifies the HRS region where the Stanford current law is missing interface-barrier physics.

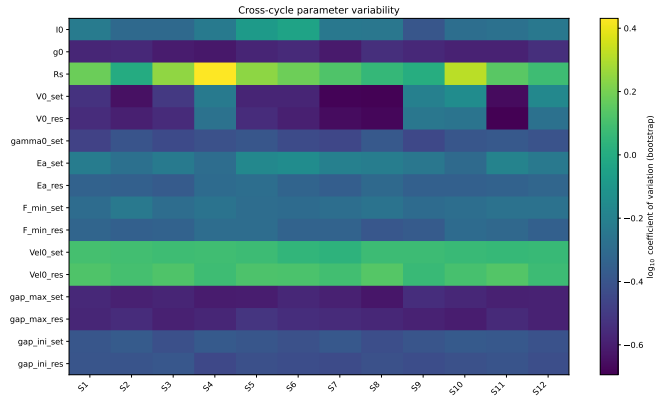


Figure 3: Bootstrap parameter variability. Low CV means repeatable extraction. Branch voltage scales are stable, while series resistance and velocity prefactors vary strongly under cycle resampling and should be treated as fit-only quantities for these DC sweeps.

Out-of-sample fit quality: model fitted to the medoid cycle, scored on every other measured cycle

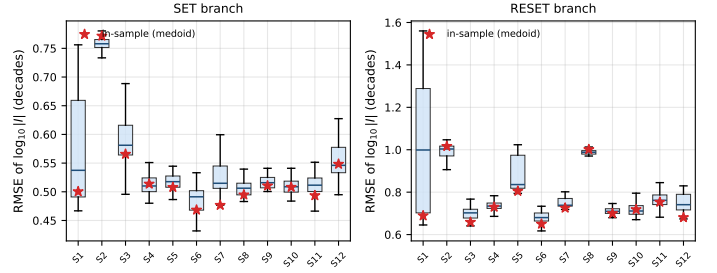


Figure 2: Out-of-sample cycle test. Boxes show the error when the medoid-cycle fit is evaluated on other measured cycles; stars show the fitted cycle. The overlap means the parameter set describes the measured cycle population rather than only the selected fitting trace.

Bootstrap-identifiable parameters vs deposition condition (error bars: cycle-bootstrap 95% CI)

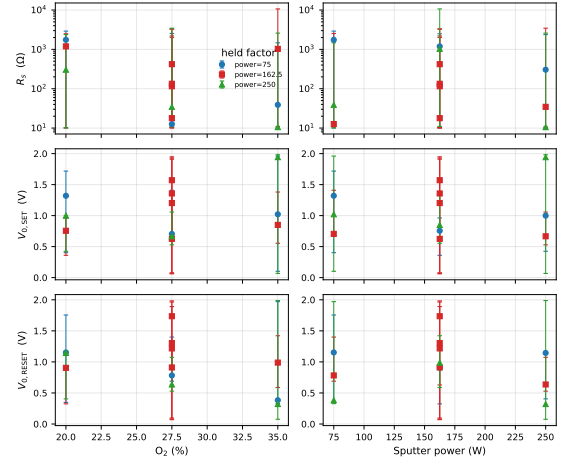


Figure 4: Deposition-resolved extraction. Branch voltage scales have tighter bootstrap intervals and carry useful deposition information. Series resistance spans orders of magnitude, so its apparent process trend is not reliable from this data set.